IMPACT OF AN ELECTRONIC HEALTH RECORD–INTEGRATED MULTIDISCIPLINARY FOOT WOUND PATHWAY IN THE EMERGENCY DEPARTMENT

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I have nothing to disclose.

I have no conflicts of interest.

Background

Burden

- 1.6M US patients/year
- High infection + amputation risks

ED Challenges

- Complex
- Multidisciplinary
- Variable care

Impact

- Delays
- Increased LOS
- High cost

Creates an opportunity for standardization at the point of ED decision-making

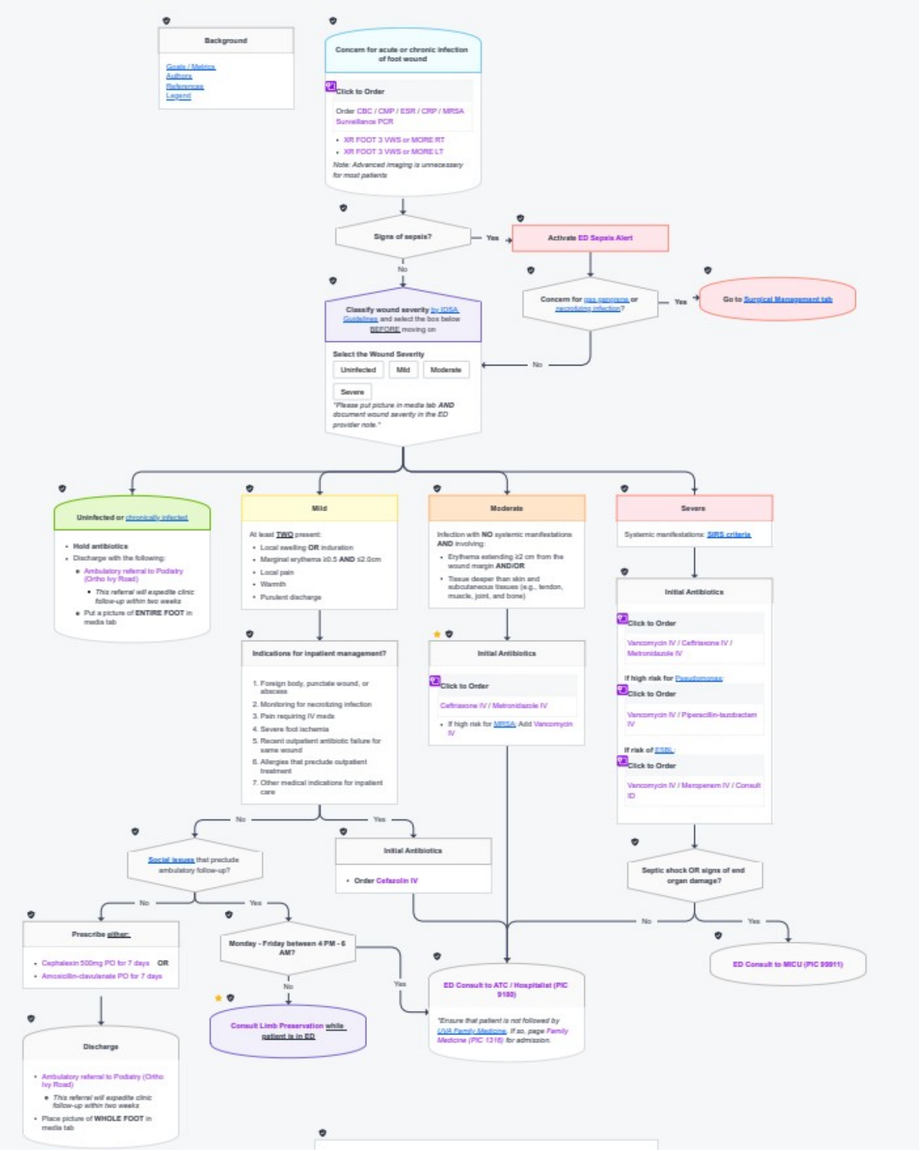
Objective

Evaluate whether a standardized limb preservation pathway reduces:

- Emergency department and hospital LOS
- Specialty consultation patterns
- Utilization of MRI

What is the Limb Preservation Pathway?

- Clinical decision support
- Standardized:
  - Diagnostic evaluation
  - Disposition
  - Early subspecialty involvement
- Later integrated into the EHR



Study Design and Population

Pre pathway
June - May 2024

Paper pathway
June - December 2024

EHR- integrated pathway
January - September 2025

- Patient encounters from the University fo Virginia Emergency Department (~75,000 ED annual visits)
- Adults ≥ 18 years
- Chief complaint of:
  - foot wounds
  - diabetic foot care
  - lower extremity infections
  - skin ulcers
  - other wound-related presentations
- ICD-10 related to foot infections AND chief complaint. match
- Exclusions: LWBS, missing coding match

- 1,287 total eligible encounters
  - 702 encounters were admitted

Study Outcomes

Primary	Secondary

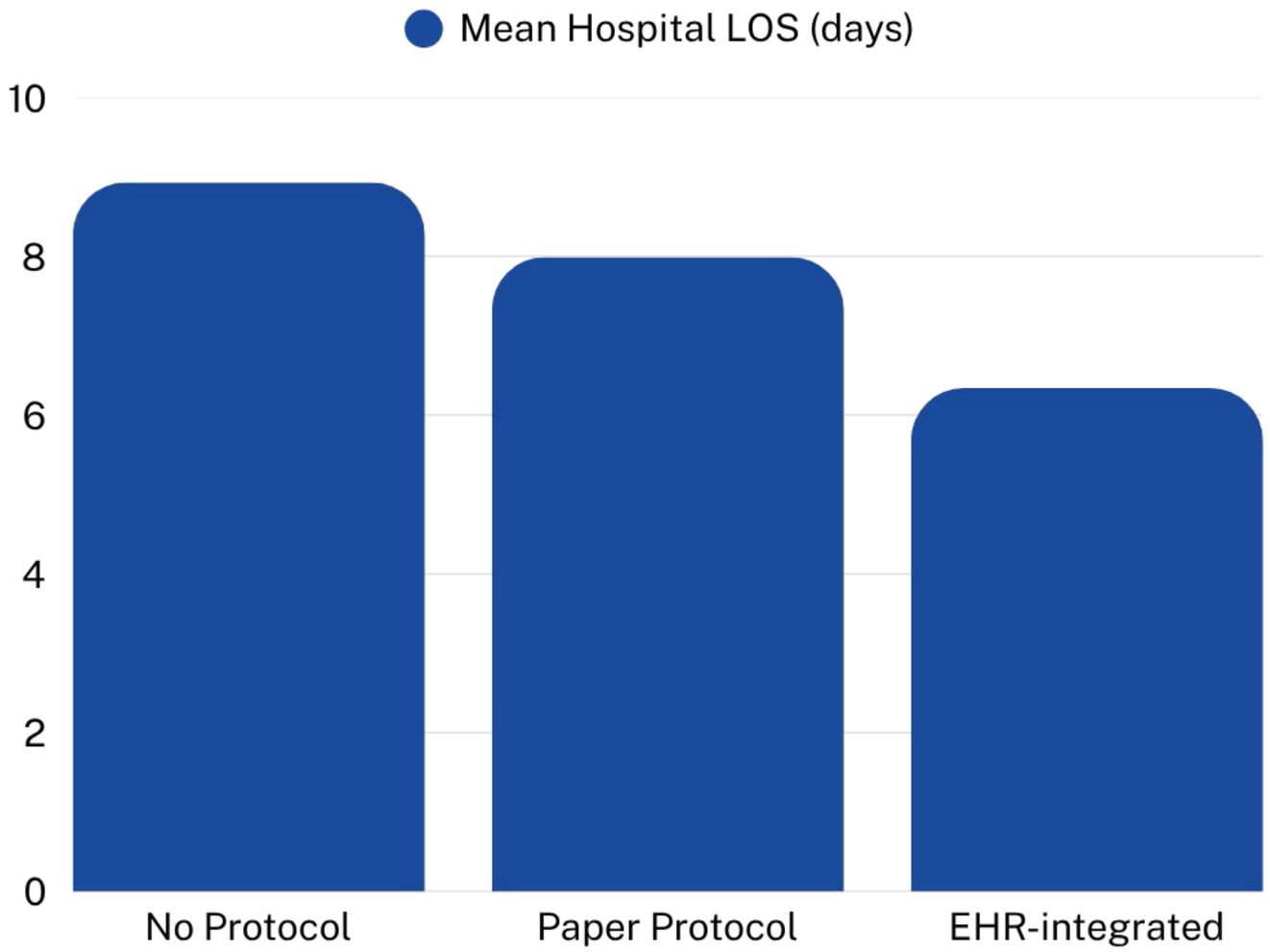
• Hospital LOS
• Admission rates
• ED LOS
	• MRI utilization
	• Consultation patterns
	  ◦ Limb preservation service
	  ◦ Orthopedic surgery

Results - Hospital LOS

No Protocol:
- Mean LOS = 8.93 days
- SD = 8.87

Paper Protocol:
- Mean LOS = 7.99 days
- SD = 10.3

EHR-integrated:
- Mean LOS = 6.34 days
- SD = 5.33



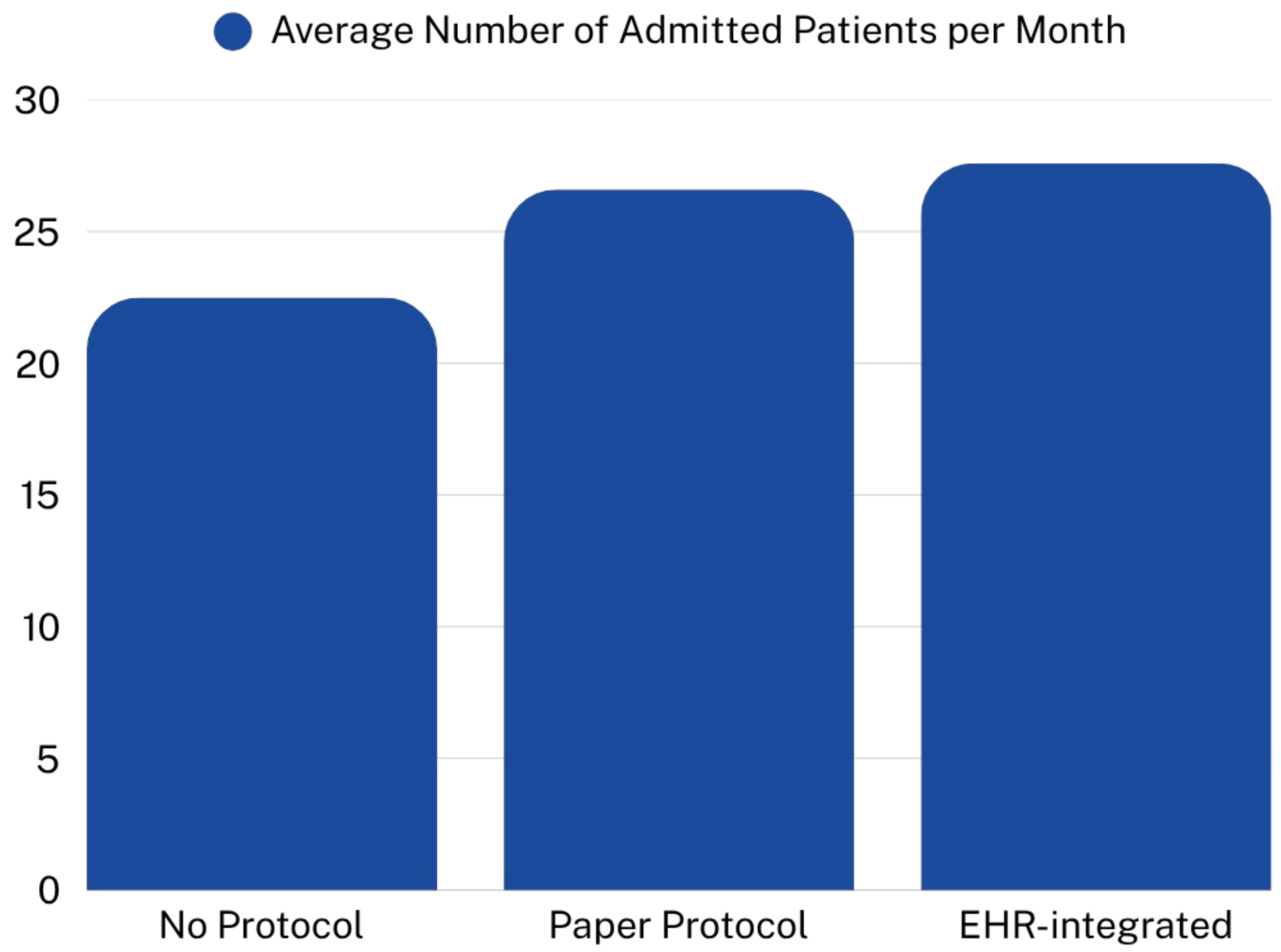
LOS decreased overall by ~2.6 days after the EHR integrated pathway was implemented

Results - Admission Rates

No Protocol:
- 270 admitted Patients
- 12 month period
- Average = 22.5 patients/month

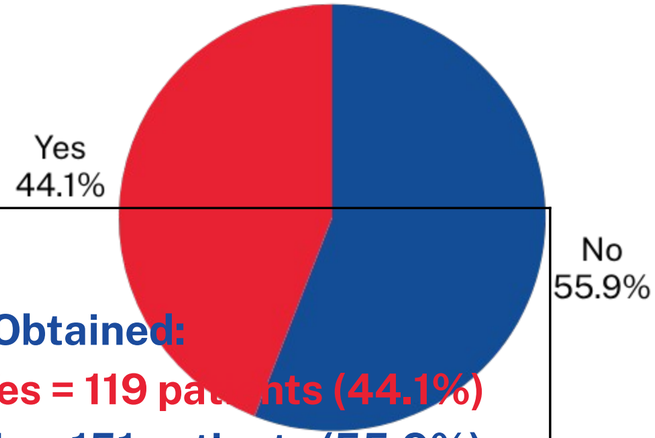
Paper Protocol:
- 239 admitted patients
- 9 month period
- Average = 26.6 patients/month

EHR-integrated:
- 193 admitted patients
- 7 month period
- Average = 27.6 patients/month



Imagi

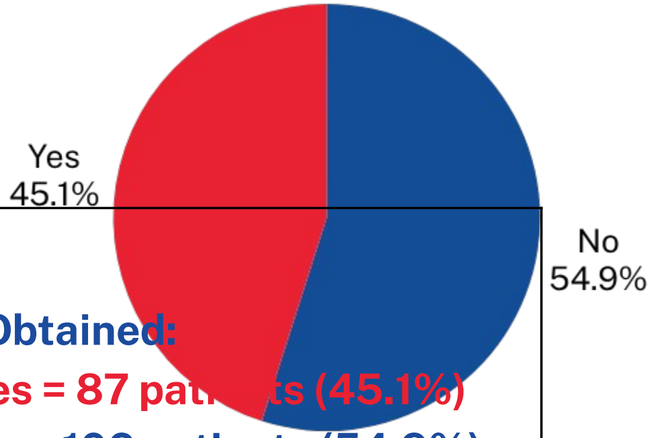
No Protocol



MRI Obtained:

- Yes = 119 patients (44.1%)
- No = 151 patients (55.9%)

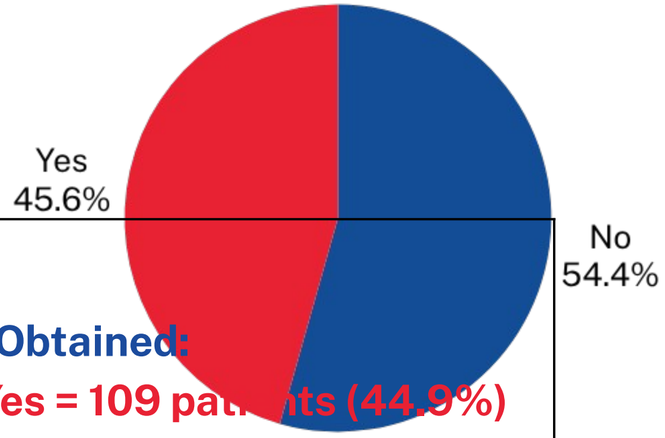
Paper Protocol



MRI Obtained:

- Yes = 87 patients (45.1%)
- No = 106 patients (54.9%)

EHR-integrated



MRI Obtained:

- Yes = 109 patients (44.9%)
- No = 130 patients (55.1%)

No changes to image ordering practices throughout pathway implementation

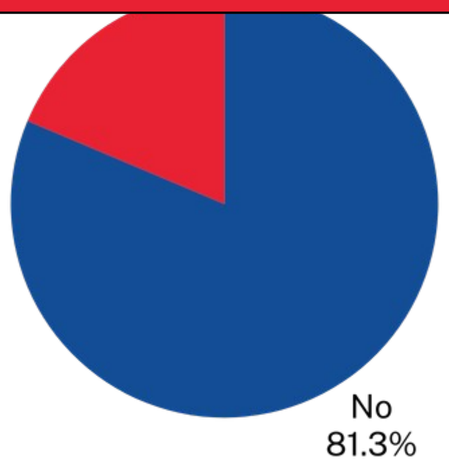
Results - Limb Preservation Consults

Was the Limb Preservation Consult used during patient care?

No Protocol

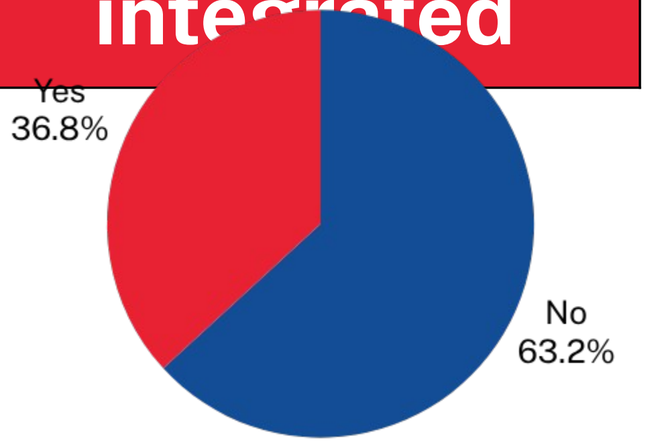
Limb preservation consultation service was not established during this phase

Paper Protocol



- Yes - 36 patients (18.7%)
- No = 157 patients (81.3%)

EHR-integrated



- Yes - 88 patients (36.8%)
- No = 151 patients (63.2%)

Key Findings

The Limb Preservation Pathway:

- Decreased hospital LOS
- Unchanged admission rates and ED LOS
- No changes in MRI ordering practice
- Increased limb preservation consultations

Why Did Hospital LOS Improve?

Improvements to:

- Diagnostic evaluation
- Standardized early decision-making
- Early specialty involvement
- Streamlined disposition pathways

Why This Matters

Impact of the Pathway

- Reduction in hospital resource burden
- Improves ED throughput
- Standardizes complex wound care
- Scalable CDS model for other conditions

Limitations

- Single-center academic ED
- ICD-10 + chief complaint identification bias
- Concurrent expansion of limb preservation service
- Lack of patient-centered outcomes (amputation, readmission, etc.)

Conclusions

The EHR-integrated pathway was associated with reduced hospital LOS and improved multidisciplinary coordination without increasing resource utilization

Embedded CDS tools for provider workflows can meaningfully improve complex care delivery

Thank you!

Questions?